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Performance analysis of deep learning architectures for plant leaves disease detection

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ABSTRACT

Plant disease is a major concern in agriculture field. Different pathogens are causes of plant disease. Manual inspection sometimes can cause errors in the disease detection over large cultivation area. Large number of resources and high level of expertise is required for correct prediction of diseases. Sometime the experts are not available in the nearby. Some diseases are so common and have same symptoms even the different experts may give different opinion about the same disease.

In this work different deep architectures are studied and tested for plant leaves disease detection. For carrying out the study plant village dataset of 20640 images are used which represents 15 class and 3 species namely pepper, potato, and tomato. Total images in the dataset are divided in three sets i.e., 70% for training, 20% for validation and 10% for testing purpose. MATLAB R2019b is utilized for experimentation. Appropriate value of hyper parameter is determined initially by varying number of epochs, learning rate. Performance of Adam optimizer is compared with Sgdm Optimizer. Finally, accuracy of eight deep learning architectures is calculated at 16 & 32 minibatch size.

1. Introduction

Indian economy is basically an agriculture-based economy. Majority of the Indian population earn their livelihood from the agriculture sector. Agriculture sector contributes a significant proportion towards the GDP of the country. Latest research and use of technology in the field of agriculture makes the country self-sufficient and a net exporter of agricultural and related goods, as opposed to being a food-begging bowl following independence. India's growing population, rising average income and the consequences of globalization will raise demand for a greater variety, quantity, and quality of food. To produce more food in greater quantity, variety, and quality put pressure on the limited amount of arable land. India is fortunate to have a lot of land that is available, 15 agro-climatic zones that, according to the ICAR, have practically every sort of weather and soil type, and the ability to cultivate a wide range of crops. Despite these facts, India's average crop productivity is still extremely poor. The population of the nation is anticipated to grow to be the largest in the world in the upcoming years, making the provision of food for all is a top concern. Use of artificial intelligence, computer vision in the field of agriculture has undergone a tremendous amount of transformation that has transformed many managerial functions into artificially intelligent systems to extract value from the growing amount of data coming from diverse sources. Deep learning, a branch of machine learning subset of artificial intelligence, has a great deal of promise to overcome many obstacles in the development of knowledge-based farming systems that will boost crop production and so resolve all problems.

2. Application of machine intelligence in agriculture

In the recent times the computer vision and machine learning algorithms have made a tremendous growth in the plant leaves detection area. Machine learning methods such as artificial neural network, support vector machine and k-nearest neighbours are employed in the field of plant leaves detection. Deep learning is an advancement of machine learning approach. The most attracting feature of deep learning over machine learning is that it is free from feature engineering. Deep neural networks basically create mapping between input which is an image and with the output in the form of crop disease pair. Deep learning architectures consist of convolution neural network. CNN consists of large

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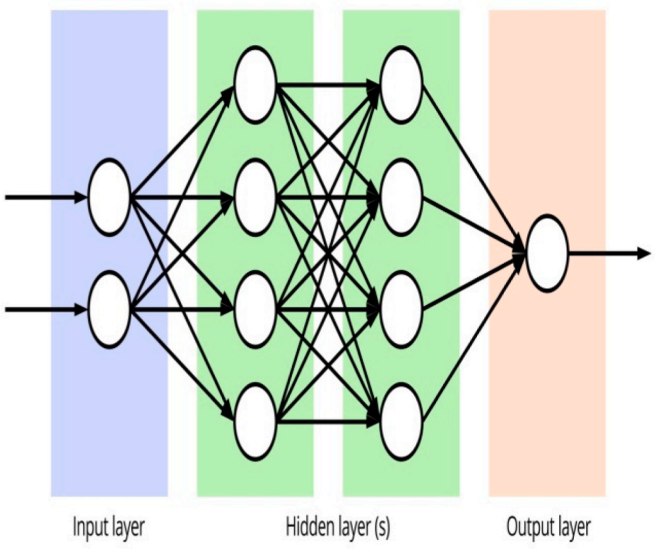


Fig. 1. Deep CNN(Google images).

number of layers such as input layer, output layer and several hidden layers as shown in Fig. 1. Several hidden layers provide greater level of accuracy as compared to shallow networks. Convolution neural networks are a class of DL methods. CNN can learn the complex problem and perform the pattern recognition when provided with a large dataset. Various DL methods now a day's find application in various fields because it facilitates the development of automated application that requires negligible amount of human intervention.

The machine intelligence-based approach like machine learning, deep learning, and use of computer vision along with application of IoT technology greatly helps the farmers and makes farming much easier. Use of drones powered by artificial intelligence (AI) is particularly beneficial for farming since it makes it simpler to monitor, scan, and analyses the crops by supplying high quality photographs. These images may be applied to machine intelligence-based algorithms for tracking the growth of crops at one hand and makes the farmer aware about the threats of any spreadable disease at the earliest. The main application area of the machine intelligence-based approaches is:

- **Crop Management:** Crop management is a crucial activity for raising crop quality. In today's agriculture, crop management functions including crop monitoring, field scanning, and so forth, drones play a critical role. Deep learning algorithms are employed in smart agriculture to track the temperature and moisture content of the crops. Farmers may also keep an eye on their fields from anywhere in the world.
- **Water Management**: For a crop to grow properly, water is necessary. Regularly inspection of moisture level in the soil is extremely important for the development of the crops. With the use of machine intelligence approaches, we can effectively manage the water level based on the crop and soil categories.
- **Soil Management**: One can quickly identify the appropriate crops and the optimum kind of fertilizer for the soil by using machine intelligence-based techniques. For more effective soil management, SVM (Support Vector Machines), Bayesian Networks, and ANN Machine learning techniques are most used techniques. Regular monitoring of pH level and soil nutrients parameters is very helpful in determining the condition of the soil because it determines the quality and quantity of the crops.
- **Weed Detection:** Weed detection and elimination are the primary issues in the agricultural sector. Weed removal used to be done manually in older times. Although it is time-consuming and expensive, many industrialized nations now deploy robots to control weeds.
- **Disease Detection:** Crop disease is a serious risk to the agriculture industry. Manual disease identification in crops is a laborious and lengthy process. With the advancement of deep learning and its architectures, it is now a quicker process.
- **Yield Prediction**: Crop production estimation using machine learning and deep learning techniques gives a fair judgment about total yield of the crops. The decision of what to cultivate and when to cultivate it is made easier for the farmers.
- **Fruit Quality and Quantity Estimation**: Computer vision has made considerable strides recently and is primarily used to recognize objects. Use of latest machine intelligence-based techniques in the large farms will be very helpful in producing good quality of fruits and in larger quantity.

3. Related works

Yanli.et al. [1] discussed that lightweight models are much faster in the identification of disease and requires less memory space. HLNet deep learning model is proposed based on lightweight CNN and for fast and efficient identification of disease. Mohammad Suleman et al. [2] discussed the identification of cotton leaf disease. A model based on the meta deep learning is proposed which is capable enough to identify the several crop diseases. Dataset contains 2385 images of healthy and diseased leave. The proposed model has achieved 98.53 accuracy. Plant image recognition using deep learning techniques based on leaf vein patterns was first developed [3]. Three species of leguminous plants—white bean, red bean, and soybean—were classified using 3–6 layers of CNN technology. Deep learning models was trained to evaluate the model's performance on the Plant Village dataset in Ref. [4]. The performance of the two well-known CNN architectures AlexNet and GoogLeNet were evaluated using the three scenarios (color, grey and segmentation), and it was determined that GoogleNet performed better than AlexNet. On the test set, the trained model had a 99.35% accuracy rate. To conduct symptom-wise recognition of four cucumber illnesses, Ma et al. [5] used a deep CNN (i.e., downy mildew, anthracnose, powdery mildew, and target leaf spots). The recognition precision was 93.4%. A system based on CNN was presented by Kawasaki et al. [6] to identify cucumber leaf disease, and it achieved an accuracy of 94.9%. Muammer et al. [7] used DL methods for recognizing the plant leaf disease and pasts. Nine Architecture of deep learning was used for feature extraction. Features were then classified by SVM, ELM and k-nearest neighbor method. Highest accuracy of 97.86 was achieved with ResNet 50 model and SVM, classifier. The main limitation of this theory was that the data set utilized for experimentation was very small. Only 1965 images of 8 different plant leaf diseases were used. A deep CNN framework was created by Ghosal et al. [8] to recognize and categorize 8 different types of soybean stress. Additionally, it will offer an explanation mechanism and make predictions using the top-K high-resolution feature maps that isolate the visual symptoms. With no comprehensive symptom annotation by experts, the unsupervised recognition of visual symptoms allowed for identification (a type of foliar stress), classification (low, medium, or high stress), and creation of a quantitative measure of stress intensity, instead of considering the entire leaf, Barbedo [9] and Lee et al. [10] suggested the usage of lesions and spots. The benefits of this approach include the ability to identify the presence of multiple diseases on the same leaf and the ability to enhance the data by segmenting the leaf picture into various sub-images. The GoogLeNet model was utilized in this study by J. Wang [11] to identify 79 diseases in 14 kinds of plants that were present in challenging field and experimental conditions. The total accuracy rate for a single lesion and area was 94%, which was higher than the 82% accuracy rate for the full image. To identify the areas affected by the wheat disease, Qiu et al. [12] employed the Mask-RCNN, whose feature extraction network was either ResNet50 or ResNet101. The average accuracy on the test dataset

was 92.01%. Based on the neural structure search technique, Huang et al. [13] approached for identifying plant leaf diseases can automatically learn the neural network's structure to the necessary depth on the PlantVillage. According to the results of the research, a suitable network structure was found on the datasets of imbalanced and balanced, and the model's recognition accuracy was 98.96% and 99.01%, respectively. However, accuracy dropped to 95.40% if the grey image balance was not corrected. In order to detect camellia leaf diseases, Long et al. [14] employed AlexNet for two types of training: training from scratch and transfer learning from the ImageNet (4 kinds of diseases and healthy). The findings demonstrated that transfer learning may considerably enhance the models' classification performance and convergence speed, with classification accuracy reaching 96.53%. Saliency maps were first used to show the signs of plant diseases by Brahimi et al. [15] using the precision (P), recall (R), F1 score, and overall accuracy metrics, Cruz et al. [16] employed segmentation and edge mapping to identify plant diseases to discussed detect olive plant diseases using the upgraded LeNet model. A new DL model network was created by Brahimi et al. [17] to identify the spots of plant illnesses, and when compared to the current plant disease treatment approaches, the new method produced a clearer visualization impact. In Ref. [18] J Arun Pandian proposed a 14-layer deep CNN model for detection of leaf disease using leaf images. Three augmentation techniques are used. Dataset contains 147500 images of 58 healthy and diseased classes and on no leaf class. The proposed DCNN model achieved 99.96555% accuracy. The proposed model gives better performance than other existing learning approaches. Waleed albattah in Ref. [19] has introduced custom CentreNet framework with DenseNet-77 as a base network. Proposed method adopted three stages strategy i.e use of annotation to extract region of interest, at second stage proposed CentreNet in which DenseNet-77 is used for key point abstraction. Finally, CentreNet is used for categorization of disease. In Ref. [20] C. Jackulin et al. carried out a review study on various ML and DL techniques in plant disease detection using AI and ML. author has also compared the performance of ML and DL techniques and their usage in various study. Dahiya et al. in Ref. [21] discussed about the deep learning architecture, various CNN related parameters. Various factors that affect the performance of DL model in plant leave disease detection is also discussed.

4. Artificial neural network

Artificial neural networks are based upon the pattern of neuron connection of the brain. These networks try to learn the things like brain. Neuron nodes in artificial neural networks are linked together like a web, just like in the brain. There are millions of neurons in the human brain. A cell body, which is the component of each neuron, oversees processing information by transporting it into and out of the brain. The neural network tries to seek pattern from the data supplied to produce desired output from the input information, which receive a variety of information based on an internal weighting system. Backpropagation which stands for backward propagation of error, is a set of learning rules that ANNs use to refine their output results, like human being learn various rules and guidelines to produce a desired outcome.

4.1. Convolutional neural network

A Convolutional Neural Network (CNN) is a special class of ANN. These are deep networks which accepts image as an input and assign some internal weights and biases to different parts or the objects appearing in the image and try to differentiate the different objects. The special feature which makes it a popular choice among the researcher is that it requires little or almost no preprocessing as compared to the traditional approaches. The pre-processing required in CNN is much lower than the other classification algorithms. The various parameters which need to be suitably defined are as follows:

Table 1
AlexNet architecture details.

Layers	No. of filters	Filter Size
1st convolutional layer	96	11 x 11
2nd convolutional layer	256	5 x 5
3rd convolutional layer	384	3 x 3
4th convolutional layer	384	3 x 3
5th convolutional layer	384	3 x 3

4.1.1. Filter size

Various filter size is available in the CNN as 1x1, 3x3 & 5x5 etc. Depending upon the application and the datasets, various filter size determines the features sets to be detected by the CNN.

4.1.2. Padding

Padding is required in the CNN so that after convolution the output of the layer is not reduced in size. Since CNN contains multiple convolution layers, hence without padding it may lose some area after several convolution steps.

4.1.3. Polling

Polling layers in the CNN is required to reduce the number of parameters for layers presents inside the CNN. It is also helpful in reducing the calculations during the learning phase.

4.1.4. Stride

Stride determines the movement of filters across the pixel of image horizontally. Stride is denoted by S.

4.1.5. Convolution arithmetic

The dimension of the output is determined by the following equations.

$$L_{q_p} = \frac{(N - F)}{S + 1} \tag{1}$$

$$L_{q_p} = \frac{(N + 2P - F)}{S} + 1 \tag{2}$$

where, N: length on side of activation layer; S: Stride; F: Filter size; P: Depth of padding.

4.1.6. Confusion matrix

A Confusion matrix determines the accuracy of model under consideration. The structure of confusion matrix is N x N. This matrix is a basically determines the performance of the model by comparing the actual class with the predicted class. Confusion matrix gives a fair representation of model performance.

4.2. Deep CNN architectures

4.2.1. AlexNet

Development of AlexNet [22] has revived the interest of the researcher towards CNN. It is a pretrained CNN.t AlexNet consists of 8 weight layers out of which are 5 layers are responsible for convolution and three layers having fully connected connection (Table 1 & Fig. 2). Three max-pooling layers are also used after the 1st, 3rd and 5th layer and computed nearly sixty million parameters. The architecture also contains some activation layer.

4096 neurons are presents in fully connected layer. The second fully connected layer is connected to a SoftMax classifier which gives the output in 1000 classes.

The performance of the AlexNet is degrades due to overfitting. Augmentation technique gives satisfactory results while dealing with overfittings. Two different types of augmentation techniques may be utilized for the above purpose, one is positional augmentation comprises

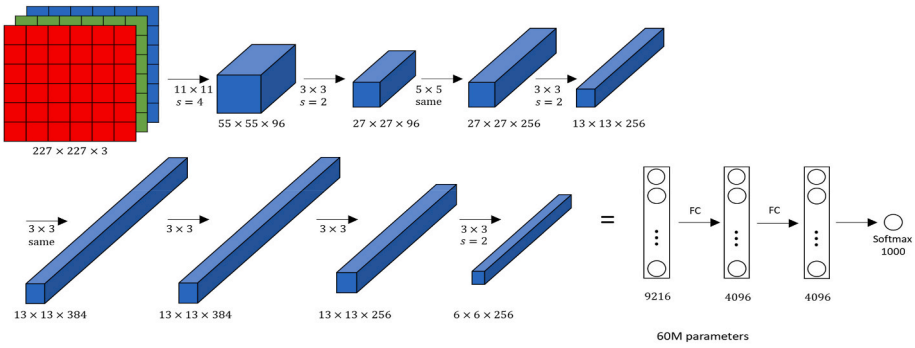


Fig. 2. AlexNet (src: Google images).

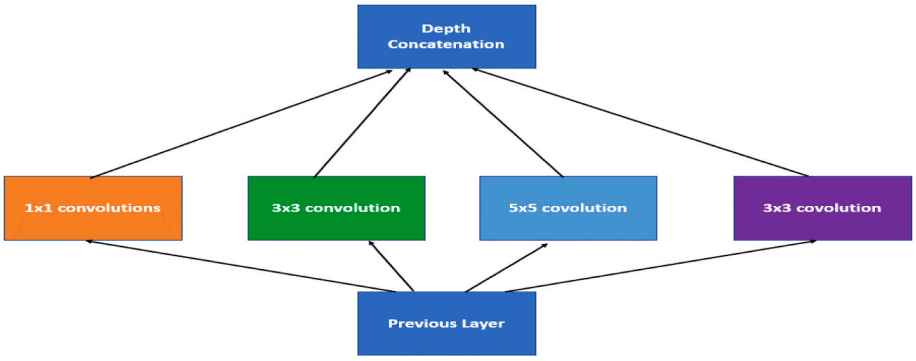


Fig. 3. Naïve inception model.

of the image's translation and reflection and the other one includes color augmentation (Brightness, hue, etc.) generates images by using differing intensities in the images. The other techniques to avoid the overfitting are to use dropout. Dropout leads to the generalization of the network. Due to dropout technique some of the units may be ignored leading to the different sample calculations and enable the model to extract more unique features/patterns.

4.2.2. *GoogleNet*

GoogleNet architecture which is also known as Inception came into existence in 2014 after AlexNet. Inception network has three versions, named Version 1, 2 and 3. The first version of the inception module was named GoogleNet [23]. GoogleNet has less error rate than other architectures. This architecture uses the features of 1x1 convolution layers and global average max pooling. These features enable it to go deeper within the architecture layers. The number of parameters in GoogleNet architecture is four million which is much less than sixty million parameters of AlexNet. GoogleNet architecture consists of 22 deep convolution layers along with 27 global average pooling layers and 9 inception modules. The global average pooling layer relates to the inception modules. Such models have repeating modules.

An inception network is a DL network that is made up of some echoing modules called inception modules. Convolution neural networks extract the features from an image at different levels and scales. The main idea is the same as the human visual cortex functions by extracting different information from different patterns; this information is accumulated at the end and form larger perceptions of objects. An inception module is made up of 1x1, 3x3 and 5x5 convolution layer, average max-pooling layer, and concatenation layer. Initially designed inception module was known as Naive inception module as shown in Fig. 3. It was the first designed inception module.

It consists of varying filter sizes by which it can learn about spatial

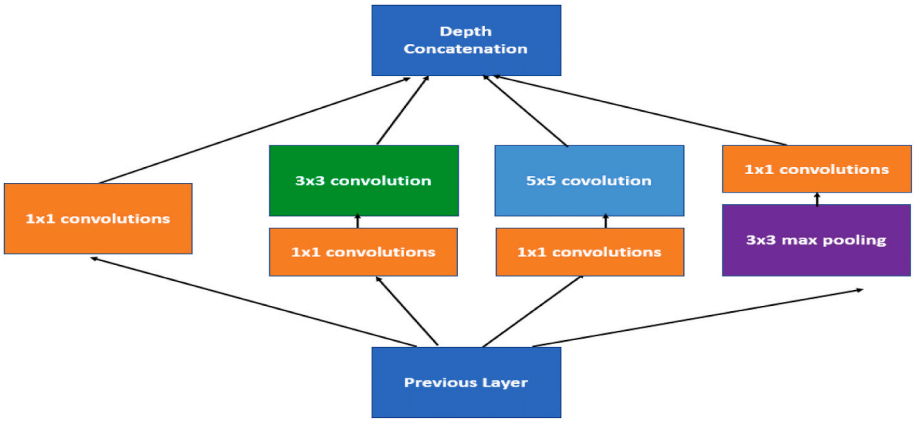


Fig. 4. Full inception model.

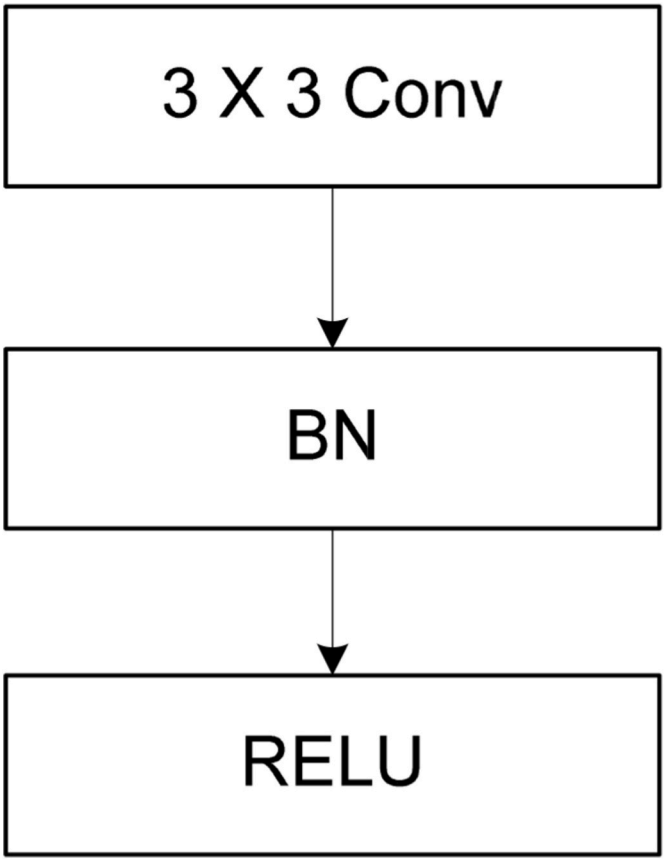


Fig. 5. Standard Convolution layer in MobileNet.

patterns at different scales. In naïve inception module an extra block of 1x1 convolution layer is added with the 3x3, 5x5, and average max pooling layers as shown in Fig. 4. It will be helpful in learning the parameters at depth and reduces dimension. The insertion of this 1x1 layer reduces the computational costs of this module. So, we can compare the computational costs of naïve and full inception modules and find that full inception module's computational costs are the times lesser. In the inception module all convolution layers work parallel and at the end output of these layers is concatenated to give final output.

4.2.3. MobileNet

Depth wise separable convolutions technique is utilized in MobileNet [24] architecture. Its architectures contain three convolutional layers. 1 × 1 point wise convolution layer is used after the initial depth convolutions. In its earlier version known as V1 the point wise convolution keeps channels the same or doubled them, whereas in V2 it tries to keep the total channels smaller. The layer performing this function is known as projection layer. It converts the high dimensional data into lower dimensions. Depth wise layer works on with 144 channels, which the projection layer may reduce the number of channels to 24 channels as it reduces the data flow across the network. The function of 1 × 1 convolution layer is to enhance the channels availability for the data before applying to the depth wise convolution. The types of convolution layer are shown Figs. 5 and 6.

4.2.4. ResNet

ResNet [25] stands for residual network. After the successful implementation of AlexNet researchers put more stress in designing the deeper network having more layers. But increasing number of.

Layers also increase the rate of training and test errors. Residual network architecture also reduces the problem of vanishing gradient. A special techniques called skip connection is used in ResNet [25], which

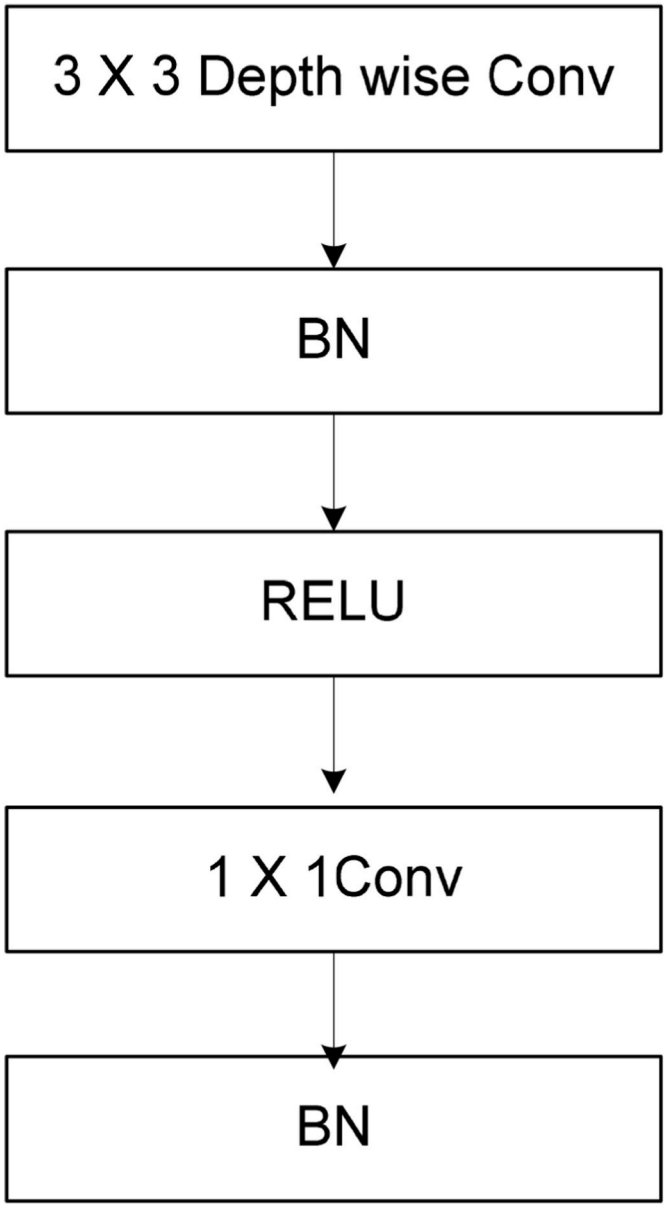


Fig. 6. Depth wise separable Convolution layer in MobileNet.

avoids connection from some of the layers and connected to the outputs. Skip connection techniques improve the performance of the network as it avoids layers which performs poor by regularization. Hence the problem of vanishing gradient can be removed. ResNet50 is a type of variant of ResNet model having 48 convolution layers along with one convolution layer and 1 average pool layer. This is a commonly used model. ResNet 101 is a deeper network having 101 layers.

4.2.5. ShuffleNet

ShuffleNet [26] a deep CNN trained on the ImageNet set and can classify the 1000 different class. The network accepts an image of size of 224x224 as an input. This network implements the pointwise convolution. It also utilizes the channel stuffing for reducing the cost factor. In the beginning the 224x224 input images is fed to the convolution layer having kernel of size 3x3 with stride of 2. This is a highly efficient network and provide good accuracy as it reduces the number of parameters that is why it also efficient in reducing the computational complexity.

Table 2
Parameters for experiments.

S. No.	Parameter	Value
1.	Mini Batch Size	64
2.	Learning Rate	0.001
3.	Epochs	30,40,50
4.	Optimizer	Adam

Table 3
GoogleNet Architecture with 30,40,50 epochs.

S. No.	Epochs	Accuracy (%)	Training Time(hrs)
1.	30	97.63	1:46:27
2.	40	97.65	3:09:48
3.	50	95.61	3:13:17

4.2.6. SqueezeNet

SqueezeNet [27] is an 18-layer deep network on ImageNet dataset. It can classify images in to 1000 classes. SqueezeNet accepts 227*227 images.

5. Experimental study

In this work, initially experiment is performed with the different set of hyperparameters. For finding the best set of hyperparameters GoogleNet architecture is used. For experimental study a version of PlantVillage dataset containing 20640 images of 15 class and 3 species namely pepper, potato, and tomato are taken. Total images are divided in three sets i.e 70% for training, 20% for validation and 10% for testing purpose. Matlab R2019b is utilized for experimentation. Six experiments are performed:

5.1. Experiment 1

5.1.1. Setting UP OF number of epochs

In this, three experiments were carried out to find out the appropriate number of epochs to calculate the accuracy of the models in minimum time. The number of epochs taken in this study are 30, 40 & 50. 64 minibatch size, learning rate 0.001 with Adam optimizer is used. Results of the study are summarized in Table 2 ad Table 3. Results showed that when the number of epochs increased from 30 to 50, there is significant impact on the accuracy of the models. But the training time is increased and almost double when we increased number of epochs from 30 to 50. The results are graphically represented in Fig. 7.

5.2. Experiment 2

5.2.1. Determination of learning rate

In this study number of epochs is set to 30 and minibatch size is set to 64. GoogleNet architecture is used with Adam optimizer. The learning rate is varied from 0.001(10^{-3}) to 0.000001(10^{-6}). Results showed that at 0.0001(10^{-4}) the highest accuracy 99.06 is achieved with learning time 1:47:21 h. At 10^{-5} and 10^{-6} learning rate the accuracy starts decreasing due to overfitting of models. At (10^{-4}) learning rate the learning time is slightly more than 10^{-3} but less than 10^{-5} and 10^{-6} learning rate. Results are summarized in Table 4 and represented in Fig. 8.

5.3. Experiment 3

5.3.1. Comparison of optimizers

During this part of experimentation Adam optimizer is compared with the sgdm optimizer with GoogleNet architecture with 30 number of epochs, 10^{-3} to 10^{-6} learning rates. The results are summarized in Table 5 and represented in Fig. 9. Results in Fig. 9 indicates that Adam optimizer is better than Sgdm optimizer since is it is not stuck over local minima. The performance of Adam optimizer is best at 10^{-4} Learning rate.

5.4. Experiment 4

5.4.1. Determination of appropriate minibatch size

In this part, GoogleNet is studied with variation in minibatch size at minibatch size values 16,32,64. Learning rate 10^{-4}, 30 epochs and adam optimizer is used. Accuracy against minibatch size is plotted in Fig. 10 and summarized in Table 6.

From the above results, it has been found that minibatch size

Table 4
Determination of learning rate with GoogleNet.

Sr. No.	Initial Learning Rate	Accuracy (%)	Learning Time (hrs)
1.	10^{-3}	97.63	1:46:27
2.	10^{-4}	99.06	1:47:21
3.	10^{-5}	97.80	2:00:12
4.	10^{-6}	90.07	2:20:45

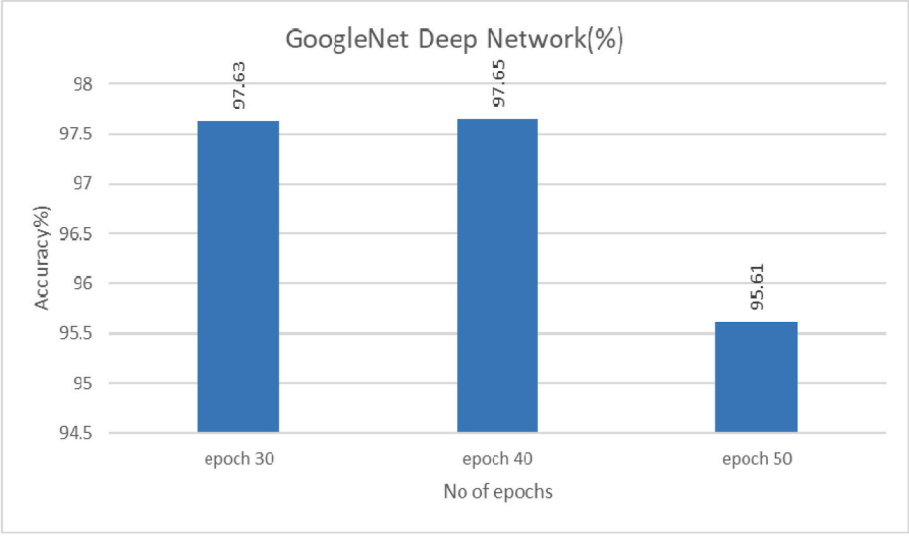


Fig. 7. Accuracy vs no. of epochs (GoogleNet).

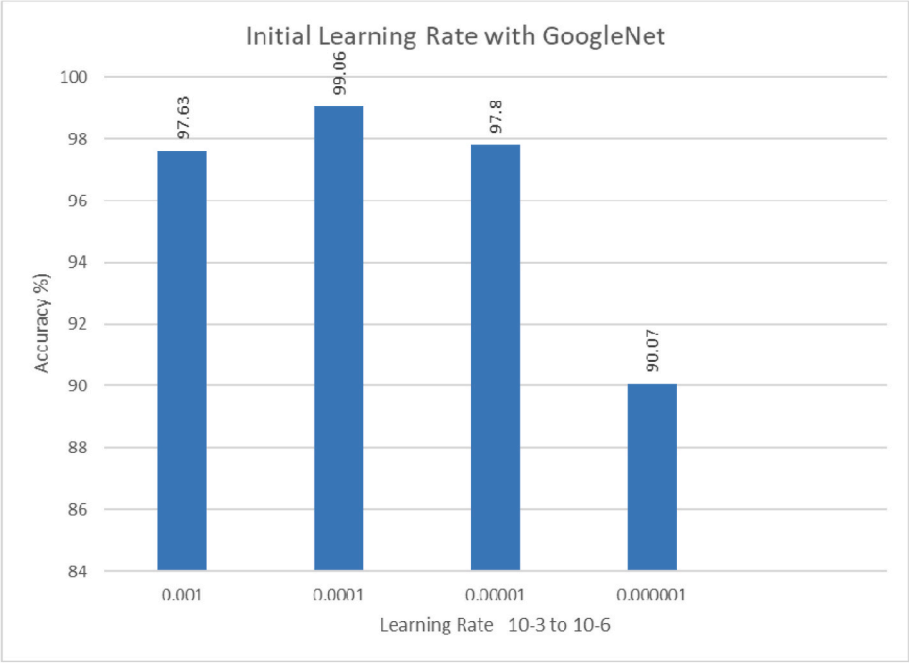


Fig. 8. Determination of learning rate with GoogleNet.

Table 5
Comparison of optimizers.

Sr. No.	Initial Learning Rate	Adam Accuracy (%)	sgdm Accuracy (%)
1.	10^{-3}	97.63	98.74
2.	10^{-4}	99.06	97.02
3.	10^{-5}	97.80	86.14
4.	10^{-6}	90.07	42.94

variation has negligible effect on the accuracy of the architecture. Variation in mini batch size increases only the learning time of the models.

5.5. Experiment 5

5.5.1. Comparison of different deep learning architecture at 32 mini batch size

Eight deep learning architectures is simulated with minibatch size 32 and accuracy is calculated as shown in Table 7 and represented in Fig. 11.

Results in Fig. 11 shows that performance of Alexnet and SqueezNet is poor than other networks. ResNet50 & ResNet101 achieves 100% accuracy whereas GoogleNet, MobileNet, ShuffleNet and ResNet18 achieves more than 99% accuracy.

5.6. Experiment 6

5.6.1. Comparison of different deep learning architecture at 16 minibatch size

Eight deep learning architecture is simulated with minibatch size 16 and accuracy is calculated as shown in Fig. 12 and summarized in Table 8. ResNet50 and ResNet101 achieves the highest accuracy whereas AlexNet and SquezeNet has achieved the lower accuracy.

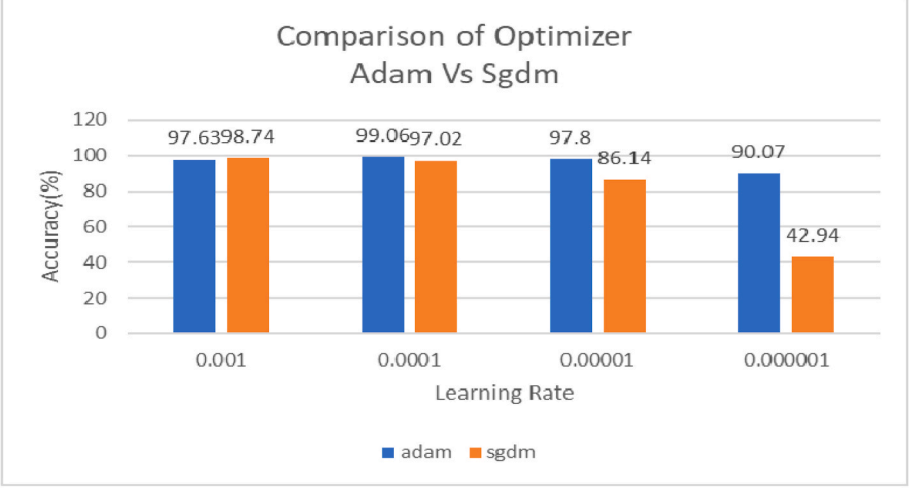


Fig. 9. Comparison of Adam and Sgdm.

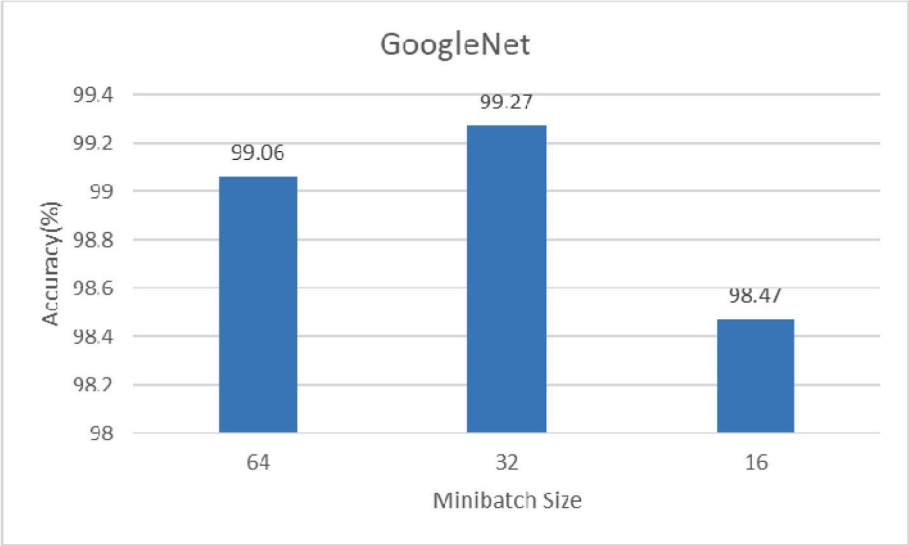


Fig. 10. GoogleNet Architecture with variation in minibatch size.

Table 6
Comparison of minibatch size.

Sr. No.	Minibatch Size	GoogleNet Accuracy (%)
1.	64	99.06
2.	32	99.27
3.	16	98.47

Table 7
Accuracy (%) at 32 minibatch size.

Sr. No.	Deep learning Architecture	Accuracy(%)
1.	GoogleNet	99.27
2.	ResNet18	99.3
3.	ResNet50	100
4.	ResNet101	100
5.	MobileNetv2	99.95
6.	ShuffleNet	99.25
7.	SqueezNet	96.15
8.	AlexNet	96.97

6. Conclusion

In this work, study is carried out for determining the appropriate hyper-parameters for comparing the different deep learning architectures for plant leaf disease detection using PlantVillage dataset. GoogleNet architecture is used for study, and it is found that 30 epochs, 0.0001 learning rate is suitable for study of plant leaf disease detection using PlantVillage dataset. Experiment results suggest that 30 numbers of epochs are sufficient as it takes lesser time for training without any significant degradation in accuracy. 10^{-4} learning rate gives better results without overfitting of the modals. Performance of Adam optimizer is found better than Sgdm optimizer. Using these hyper parameters eight different architectures GoogleNet, ResNet18, ResNet50, ResNet101, MobileNetv2, ShuffleNet, AlexNet and SqueezNet is analyzed, and it is found that ResNet50 & ResNet101 performs better than other networks whereas the performance of Alexnet and SqueezNet is poorer as compared to other network at 32 & 16 Mini batch size.

CRediT authorship contribution statement

Sachin Dahiya: Writing – original draft, Validation, Reviewing and

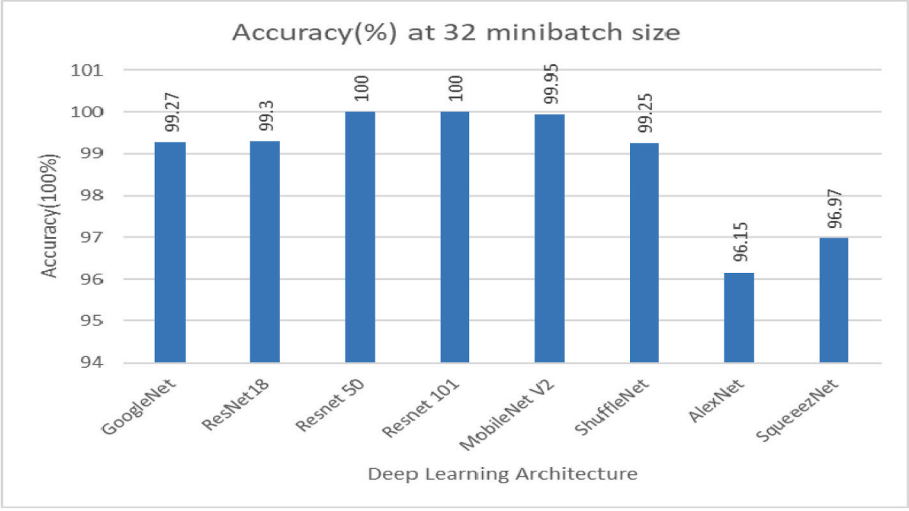


Fig. 11. Accuracy (%) at 32 minibatch size.

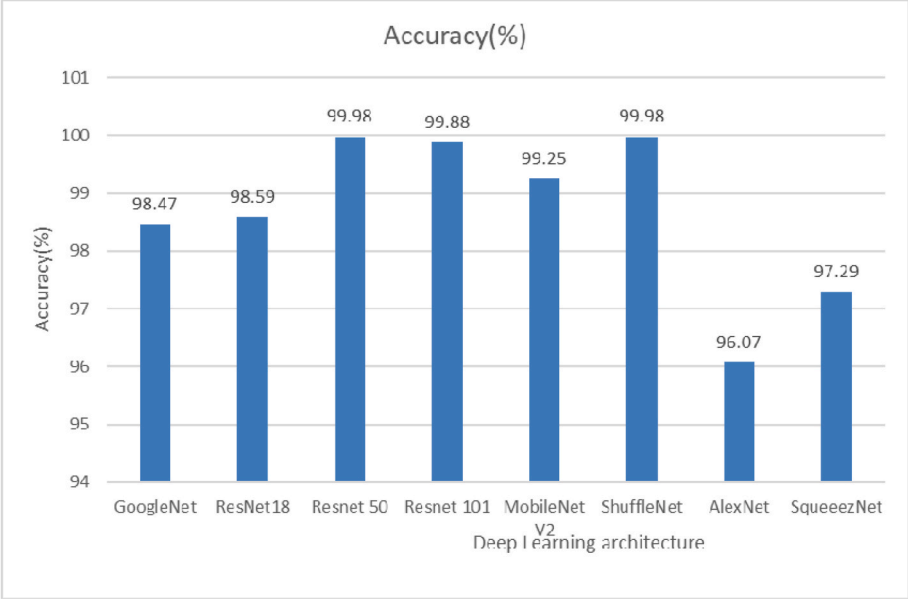


Fig. 12. Accuracy graph at 16 minibatch size.

Table 8
Accuracy at 16 minibatch size.

Sr. No.	Deep learning Architecture	Accuracy(%)
1.	GoogleNet	98.47
2.	ResNet18	98.59
3.	ResNet50	99.98
4.	ResNet101	99.88
5.	MobileNetv2	99.25
6.	ShuffleNet	99.98
7.	SqueezNet	97.29
8.	AlexNet	96.07

draft and Editing, Dr. **Tarun Gulati:** Supervision, **Dr. Dushyant Gupta:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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